Subject:
Physics

System:
Light and Sound

Scientists study natural disasters, such as tornados and asteroid impacts, using various simulation methods. Impact damage from tornado debris (eg, lumber, bricks, tree limbs) or asteroids is measured by propelling objects at high velocities using devices like an air cannon.

Sensors monitor the object's location and speed during testing. For example, the light source and photoresistor shown in Figure 1 can detect when the projectile passes between the source and the photoresistor.

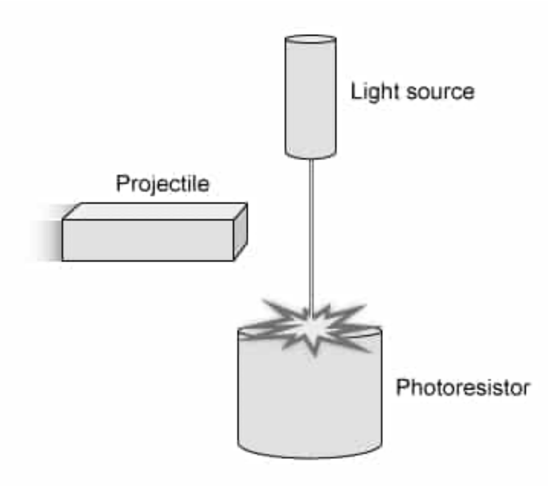


Figure 1 Schematic of a light source and photoresistor for monitoring the position of a projectile

The electrical resistance of a photoresistor changes depending on the intensity of light. Figure 2 shows that increasing the light intensity, measured in units of lux, decreases the resistance of the photoresistor.

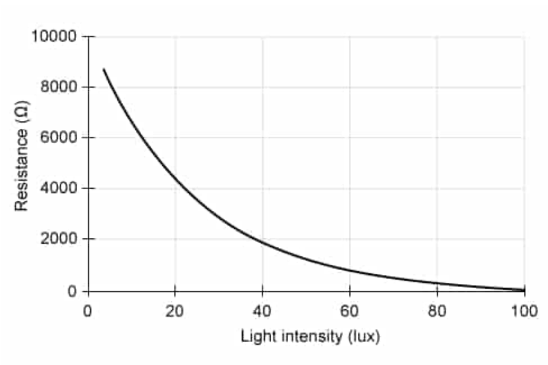


Figure 2 Electrical resistance of a photoresistor exposed to different light intensities

Photoresistors are designed to be sensitive to a specific wavelength of light, such as ultraviolet light, visible light, or infrared light. Without exposure to light, the electrons in the photoresistor are bound to the atomic structure and cannot contribute to the flow of electric current, resulting in a high resistance. When exposed to photons of light with energies greater than the binding energy, the electrons are ejected from the atoms and can flow as current, decreasing the resistance of the material. Light-filtering materials are used to limit activation of the photoresistor to only a narrow range of energies above its binding energy.

Table 1 Binding Energies of Photoresistors Sensitive to Specific Wavelengths of Light

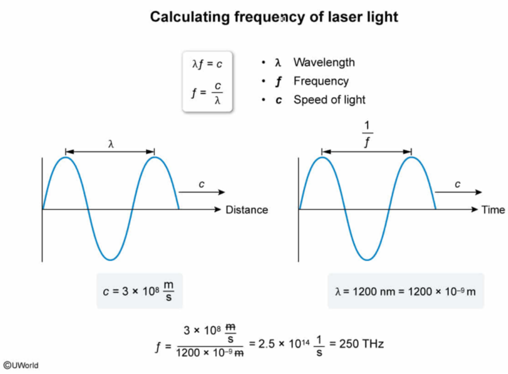
Photoresistor range	Binding energy (eV)
Ultraviolet light	10
Visible light	2.0
Infrared light	0.1

The speed of an asteroid is monitored using a 1200 nm laser. What is the frequency of the light emitted from this laser? (Note: Speed of light c = 3.0 × 10^8 m/s.)

- A. 250 THz [63%]
- B. 40 THz [11%]
- C. 3.6 THz [10%]
- D. 250 GHz [13%]

Correct
63% answered correctly

Explanation:



The product of the wavelength λ and the frequency f of electromagnetic radiation equals the **speed of light** c:

$$\lambda\ f = c$$

Rearranging this equation to solve for f yields:

$$f = \frac{c}{\lambda}$$

In this question, the λ of the laser light is 1200 nm. Because "nm" represents units of nanometers, λ equals 1,200 × 10^{-9} m. Hence, f of the laser is calculated from the equation above:

$$f = \frac{3 \times 10^8\ \frac{\cancel{\text{m}}}{\text{s}}}{1{,}200 \times 10^{-9}\ \cancel{\text{m}}} = \frac{3}{1{,}200} \times 10^{8+9}\ \frac{1}{\text{s}}$$

$$f = 0.0025 \times 10^{17}\ \frac{1}{\text{s}} = 250 \times 10^{12}\ \text{Hz}$$

Because 10^{12} Hz equals 1 THz, the value of *f* can be expressed as:

$$f = 250\ \mathrm{THz}$$

Therefore, *f* of the laser light equals 250 THz.

(Choice B) 40 THz is obtained by calculating the ratio of the wavelength and the speed of light, and incorrectly calculating the ratio of the exponents.

(Choice C) 3.6 THz is calculated from the product of the wavelength and the speed of light, and incorrectly using 3 m/s for the speed of light.

(Choice D) 250 GHz results from incorrectly using the value $1{,}200 \times 10^{-6}$ m for the wavelength.

Educational objective:
The frequency of electromagnetic radiation equals the ratio of the speed of light and its wavelength.

Time spent: 0 sec
Subject: Physics

QID:403618
System: Light and Sound

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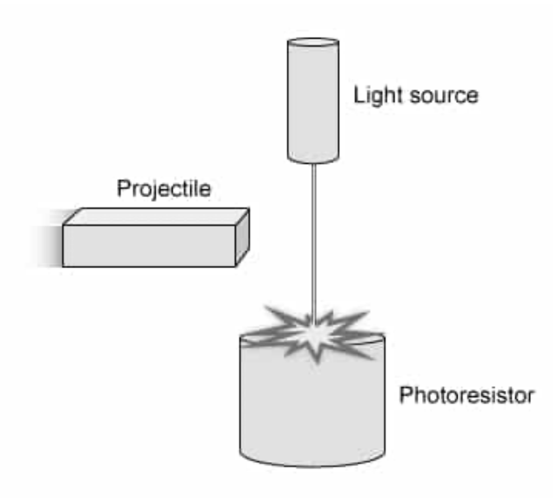

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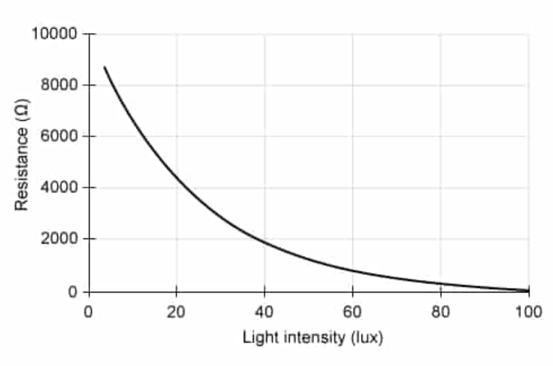

Figure 2 Electrical resistance of a photoresistor exposed to different light intensities

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Table 1 Binding Energies of Photoresistors Sensitive to Specific Wavelengths of Light

Photoresistor range	Binding energy (eV)
Ultraviolet light	10
Visible light	2.0
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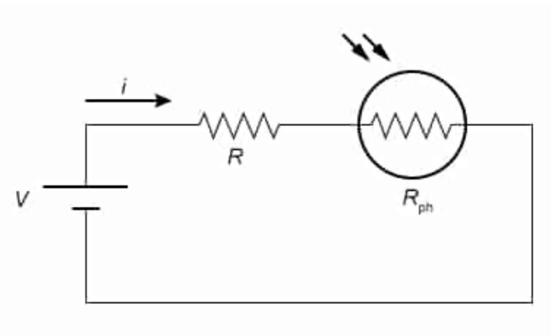
The circuit below is designed to measure light using a voltage source *V*, a fixed resistor *R*, and a photoresistor R_{ph}.

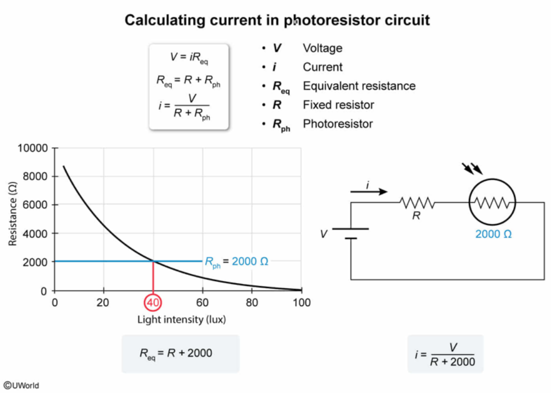


Which relationship gives the value of the current *i* when the light intensity is 40 lux?

- A. $i = \dfrac{V}{2{,}000}$ [23%]
- ✔ B. $i = \dfrac{V}{R + 2{,}000}$ [65%]
- C. $i = \dfrac{V}{40}$ [2%]
- D. $i = \dfrac{V}{R + 40}$ [8%]

Correct
65% answered correctly

Explanation:



Ohm's law states that the **voltage** *V* across a resistor equals the product of the **current** *i* and the **resistance** *R*:

$$V = \mathrm{i}R$$

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This equation can be rearranged to solve for *i*:

$$i = \frac{V}{R}$$

Furthermore, a circuit containing more than one resistor can be simplified by calculating the **equivalent resistance** R_{eq} of the overall circuit. Specifically, R_{eq} of two resistors, R_1 and R_2, connected in series is equal to the sum of R_1 and R_2:

$$R_{\text{eq}} = R_1 + R_2$$

Therefore, *i* in a circuit with two resistors connected in series equals the ratio of *V* and R_{eq}:

$$i = \frac{V}{R_{\text{eq}}} = \frac{V}{R_1 + R_2}$$

In this question, the photoresistor R_{ph} is connected in series with a resistor of fixed resistance *R*. R_{eq} of the overall circuit equals the sum of *R* and R_{ph}:

$$R_{\text{eq}} = R + R_{\text{ph}}$$

The value of R_{ph} depends on the light intensity. According to Figure 2, a light intensity of 40 lux causes the value of R_{ph} to be 2,000 Ω. Therefore, R_{eq} of the overall circuit equals:

$$R_{\text{eq}} = R + 2{,}000$$

Substituting this equation into the equation for *i* from above yields:

$$i = \frac{V}{R + 2{,}000}$$

(Choice A) This equation only considers the resistance of the photoresistor (2,000 Ω), but the current depends on the sum of the resistance of both resistors.

(Choices C and D) These equations use the intensity of the light (40 lux), but the current depends on the resistance of the photoresistor (2,000 Ω).

Educational objective:
The current in a circuit equals the ratio of the voltage and the resistance. The equivalent resistance of a circuit consisting of two resistors in series is the sum of the two resistors.

Time spent: 0 sec
Subject: Physics
QID: 403619
System: Light and Sound

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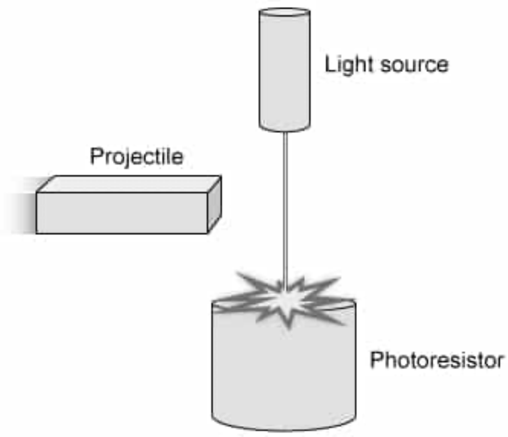

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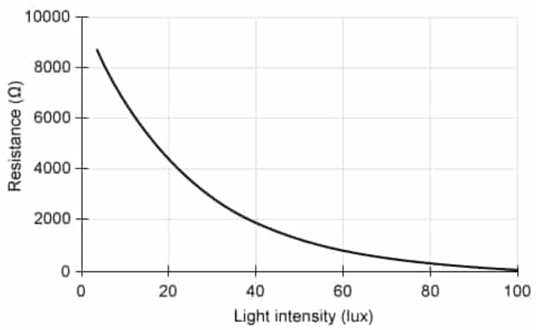

Figure 2 Electrical resistance of a photoresistor exposed to different light intensities

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Table 1 Binding Energies of Photoresistors Sensitive to Specific Wavelengths of Light

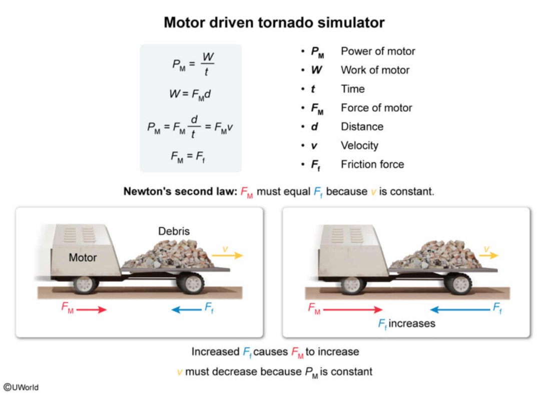
Photoresistor range	Binding energy (eV)
Ultraviolet light	10
Visible light	2.0
Infrared light	0.1

A tornado simulator uses a motor with a fixed power output to drive a dolly and propel debris horizontally toward a structure at constant speed. Which of the following will occur if the friction force on the dolly's wheels is increased?

- A. The force applied by the motor will remain constant [9%]
- B. The force applied by the motor will decrease [9%]
- C. The speed of the dolly will remain constant [4%]
- ✔ D. The speed of the dolly will decrease [76%]

Correct
76% answered correctly

Explanation:



Mechanical power *P* is defined as the ratio of work *W* and time *t*:

$$P = \frac{W}{t}$$

W can be calculated as the product of the constant applied **force** *F* and the distance *d* over which it acts:

$$W = \mathrm{Fd}$$

Combining these equations and considering that **velocity** *v* equals the ratio of *d* and *t* yields:

$$P = F\frac{d}{t} = \mathrm{Fv}$$

In this question, the power P_M of the motor is a fixed value based on its mechanical properties and is equal to the product of the force F_M produced by the motor and *v*:

$$P_M = F_M v$$

Furthermore, because the dolly always travels at constant speed (ie, has zero acceleration), Newton's second law of motion requires the two horizontal forces, F_M and the friction force F_f, to be balanced and sum to zero:

$$F_M = F_f$$

To satisfy the equation above, an increase in F_f would cause the value of F_M to also increase for the forces to remain balanced. Consequently, v would decrease because its product with F_M must equal the fixed value of P_M.

Therefore, increasing F_f leads to a proportionate decrease in v.

(Choices A and B) The force applied by the motor must balance the friction force on the dolly's wheels to maintain a constant speed. Increasing the friction force of the wheels causes the force from the motor to increase, not remain constant or decrease.

(Choice C) The power of the motor is a constant value equal to the product of the force and the velocity. Increasing the force causes the velocity to decrease, not remain constant.

Educational objective:
Mechanical power equals the product of the applied force and the velocity. For a constant value of power, an increase in the required force causes the velocity to decrease.

Time spent: 0 sec	**QID:** 403620
Subject: Physics	**System:** Light and Sound

Scientists study natural disasters, such as tornados and asteroid impacts, using various simulation methods. Impact damage from tornado debris (eg, lumber, bricks, tree limbs) or asteroids is measured by propelling objects at high velocities using devices like an air cannon.

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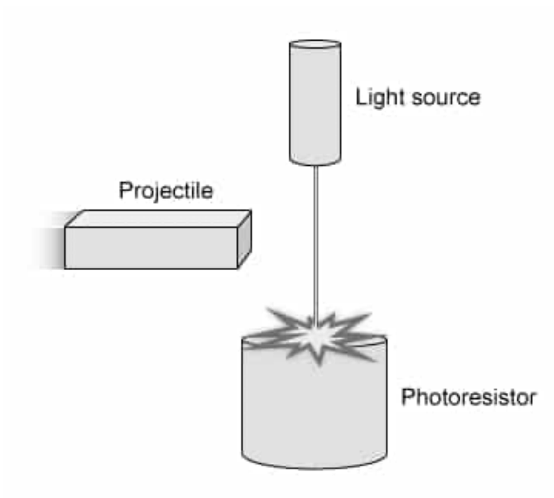

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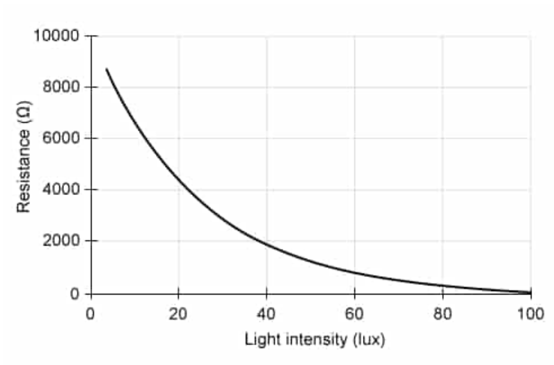

Figure 2 Electrical resistance of a photoresistor exposed to different light intensities

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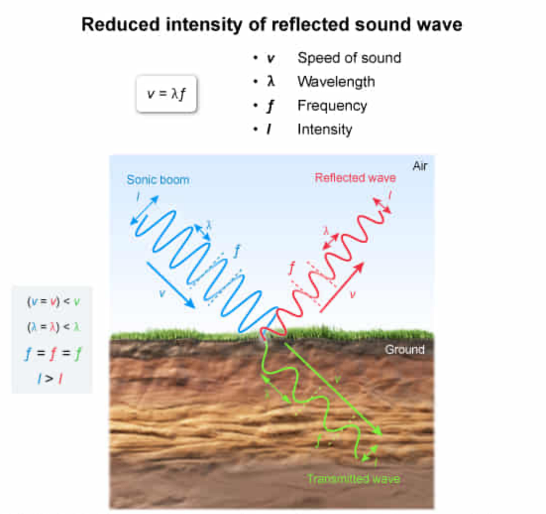
Photoresistor range	Binding energy (eV)
Ultraviolet light	10
Visible light	2.0
Infrared light	0.1

An asteroid enters Earth's atmosphere at a shallow angle and creates a sonic boom that reflects off Earth's surface. Which of the following properties of the sonic boom decreases after it is reflected off the surface?

- A. Frequency [15%]
- B. Wavelength [12%]
- C. Intensity [56%]
- D. Speed [15%]

Correct
56% answered correctly

Explanation:



Intensity of reflected sound wave decreases, but speed, wavelength and frequency remain the same.

Sound travels through a medium with a speed *v* equal to the product of the wavelength λ and the frequency *f*:

$$v = \lambda \ f$$

When sound encounters an **interface** between two media with different values for *v*, a portion of the sound wave is **transmitted** through the interface and a portion is **reflected** back into the original medium. When sound travels into the new medium, λ changes to satisfy the equation above but *f* remains constant.

The reflected sound wave has the same *v*, λ, and *f* as the original sound wave because it is still traveling in the original medium. However, the intensity of the reflected sound wave is not the same as that of the original sound wave. To satisfy the principle of conservation of energy, the energy of the original wave is divided between the reflected wave and the transmitted wave. Consequently, the energy of the reflected wave is less than the energy of the original wave.

In this question, the v, λ, and f of the reflected sonic boom are identical to those of the original sonic boom because both the original sound wave and the reflected sound wave are traveling through the air. However, the intensity of the reflected sonic boom will be lower than the original because a portion of the sound is transmitted into the ground.

(Choices A, B, and D) The speed, wavelength, and frequency of the reflected sound wave are the same as that of the original sonic boom because both waves travel through the air.

Educational objective:
When sound reflects off an interface between two media, the reflection has a reduced intensity because some of the sound is transmitted into the new media. The speed, wavelength, and frequency of the reflection are the same as that of the original sound wave.

Time spent:0 sec
Subject:
Physics

QID:403621
System:
Light and Sound

Scientists study natural disasters, such as tornados and asteroid impacts, using various simulation methods. Impact damage from tornado debris (eg, lumber, bricks, tree limbs) or asteroids is measured by propelling objects at high velocities using devices like an air cannon.

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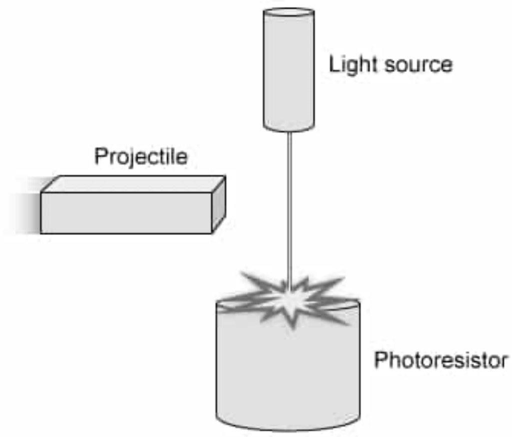

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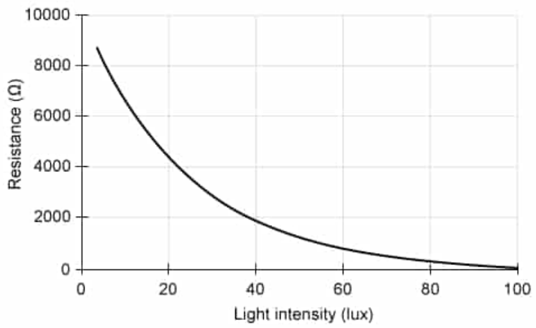

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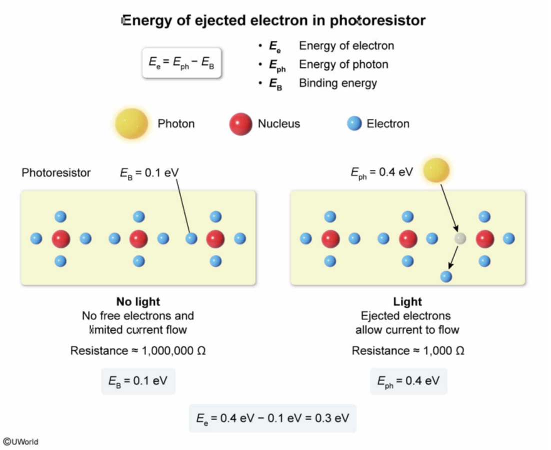
Photoresistor range	Binding energy (eV)
Ultraviolet light	10
Visible light	2.0
Infrared light	0.1

A 0.4 eV photon is absorbed by an infrared photoresistor. What is the energy of the ejected electron?

- A. 0.1 eV [10%]
- ✔ B. 0.3 eV [56%]
- C. 0.4 eV [25%]
- D. 0.5 eV [7%]

Correct
55% answered correctly

Explanation:



Light consists of **photons** with an energy E_{ph} equal to the product of its frequency *f* and Planck's constant *h*:

$$E_{\text{ph}} = \text{fh}$$

Photons can be absorbed by some materials and their energy transferred to **electrons** in the atoms. The energy required to free these electrons is the **binding energy** E_B. As a result, electrons are only ejected from an atom when E_{ph} is greater than E_B. The energy E_e of the ejected electron is equal to the difference between E_{ph} and E_B:

$$E_{\text{e}} = E_{\text{ph}} - E_{\text{B}}$$

In this question, a 0.4 eV photon is absorbed by an infrared photoresistor. Table 1 shows that E_B for an infrared photoresistor is 0.1 eV. Accordingly, E_e equals the difference between these values:

$$E_{\text{e}} = 0.4\ \text{eV} - 0.1\ \text{eV} = 0.3\ \text{eV}$$

(Choice A) 0.1 eV is the binding energy of the electron, which holds it within the atom. The energy of the ejected electron is the difference between the photon's energy and the electron's binding energy.

(Choice C) 0.4 eV is the energy of the infrared photon. The energy of the ejected electron is the difference between this energy and the binding energy.

(Choice D) 0.5 eV equals the sum of the photon's energy and the electron's binding energy. The energy of the ejected electron is the difference between these two energies.

Educational objective:
Photons can be absorbed within a material and their energy transferred to bound electrons. The energy required to eject an electron is the binding energy. The energy of the ejected electron is the difference between the binding energy and the photon's energy.

Time spent:0 sec
Subject:
Physics

QID:403622
System:
Light and Sound

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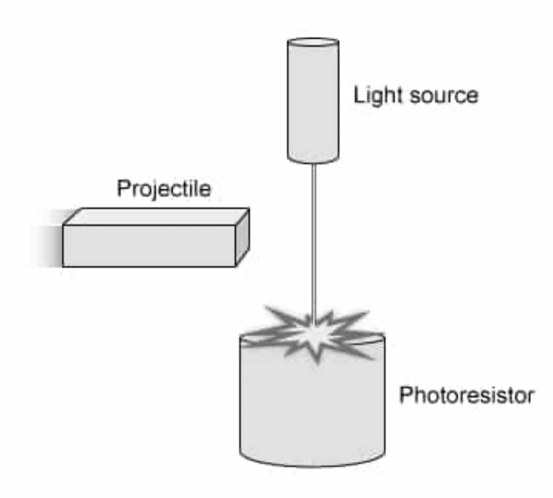

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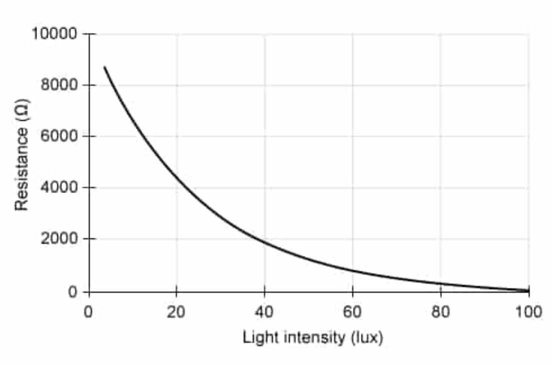

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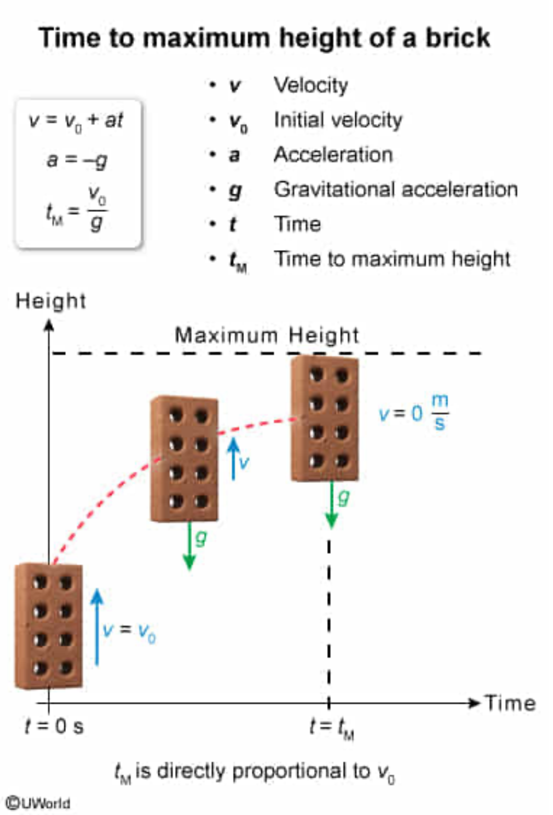
An air cannon fires a brick straight upward. The initial velocity of the brick and time required to reach its maximum height are recorded. Which of the following statements describe the expected relationship between the time to maximum height and the initial velocity? Assume that the effects of air resistance are negligible.

- A. Time to maximum height is independent of initial velocity [6%]
- B. Time to maximum height is inversely proportional to initial velocity [27%]
- C. Time to maximum height is directly proportional to initial velocity [38%]
- D. Time to maximum height is directly proportional to the square of initial velocity [28%]

Correct
38% answered correctly

Explanation:



The translational motion equations relate kinematic parameters such as acceleration a, **velocity** v, and **time** t. Specifically, an object's v is equal to the sum of its initial velocity v_0 and the product of a and t:

$$v = v_0 + at$$

In this question, a brick travels straight up into the air. The brick's *a* is constant and equal to the gravitational acceleration *g*, which acts in the downward direction (ie, negative) and causes the brick to slow down. The brick's *v* is zero when it reaches its maximum height, and it then begins to fall back to earth. Hence, the equation above can be solved for the time to maximum height t_M by setting *v* equal to zero and *a* equal to *g*:

$$0 = v_0 - gt$$

$$t_M = \frac{v_0}{g}$$

Therefore, the time to maximum height t_M is directly proportional to v_0.

(Choices A and B) The time to maximum height would not be independent of initial velocity nor inversely proportional to initial velocity because it is expected to increase when the initial velocity increases.

(Choice D) The time to maximum height would not be proportional to the square of initial velocity because the time is expected to increase proportionally to an increase in initial velocity.

Educational objective:
An object traveling straight up experiences deceleration due to Earth's downward gravitational acceleration. An object with a greater initial velocity will travel farther upward and take a longer amount of time to reach its maximum height.

Time spent:0 sec
Subject:
Physics

QID:403623
System:
Light and Sound